F70LA Life Insurance Mathematics A (2024/25)
Tutorial 7 Solutions

1. The insurer will calculate the premium for a term or whole life insurance policy assuming that the policyholder is in relatively good health; otherwise, if the insurer assumed that all purchasers were unhealthy, the cost of insurance would be prohibitive to those customers who are healthy. The assumption then is that claims will be relatively rare in the first few years of insurance, especially since most policies are sold to lives in their 30s and 40s.

   This means that the price is too low for a life who is very unwell, for whom the risk of a claim shortly after purchase might be 10 or 100 times greater than for a healthy life. The insurer therefore needs evidence that the purchaser is in good health, to avoid the risk that insurance is bought too cheaply by lives who have a much higher probability of claim.

   The objective of underwriting is to produce a relatively homogeneous insured population when policies are issued. The risk that the policyholder purchases the insurance because they are aware that their individual risk is greater than that of the insured population used to calculate the premium, is an example of *adverse selection* risk. Underwriting is a way of reducing the impact of adverse selection for life insurance.

   Adverse selection for an annuity purchaser works in the other direction – a life might buy an annuity if they considered their mortality was lighter than the general population. But, since adverse selection is likely to affect all lives purchasing annuities, more or less, it does not generate heterogeneity, and the impact can be managed by assuming lower overall mortality rates for annuitants.

2. (a)
   - $\bar{a}_{\overline{T_x}|}$ is the PV at age x of continuous payments at rate £1 per year for the remainder of the lifetime of (x).
   - $\bar{a}_{\overline{\min\{n,T_x\}}|}$ is the PV at age x of continuous payments at rate £1 per year for the remainder of the lifetime of (x) or n years, whichever is shorter.

   The difference is the PV at age x of payments from age $x+n$ onwards, while (x) is still alive. This is the PV of an n-year deferred annuity.

   (b)

$$\begin{aligned}\mathbb{E}(Y) = \mathbb{E}\left(\bar{a}_{\overline{T_x}|}\right) - \mathbb{E}\left(\bar{a}_{\overline{\min\{n,T_x\}}|}\right) =& \mathbb{E}\left(\frac{1-v^{T_x}}{\delta}\right) - \mathbb{E}\left(\frac{1-v^{\min\{n,T_x\}}}{\delta}\right)\\ =& \frac{1-\mathbb{E}\left(v^{T_x}\right)}{\delta} - \frac{1-\mathbb{E}\left(v^{\min\{n,T_x\}}\right)}{\delta}\\ =& \frac{-\bar{A}_x + \bar{A}_{x:\overline{n}|}}{\delta} = \frac{\bar{A}_{x:\overline{n}|} - \bar{A}_x}{\delta}.\end{aligned}$$

3. (a) The benefit is an annuity of 200 units per year payable continuously to (x) for at most 20 years, with payments being made only when (x) is alive. The EPV is $200\bar{a}_{x:\overline{20}|}$.

   (b) The benefit is an annuity of 500 units per year payable at the end of each year for 10 years regardless of whether (x) is alive or dead (i.e. the payment is guaranteed for the first 10 years). If (x) is alive at time 10 years, the payment of 500 units per annum continues until (x) dies. The EPV is $500\,(a_{\overline{10}|} + {}_{10}p_x\, v^{10}\, a_{x+10})$.
   This type of guaranteed payment period is a common option in life annuity contracts and is also seen in many UK defined benefit pension schemes' retirement benefits.

4. (a) The *Family Income Benefit* pays out if (x) dies within the first n years. Suppose the annuity is bought at time 0 and time is measured in years. Upon the death of (x) before time n years, a payment of 1 unit per year is paid continuously from the time of death, at time T_x, until time n. There is no payment if (x) survives to time n.

   (b) A payment is made only at time $t \le n$ if (x) has died on or before time t. The latter event has probability ${}_tq_x$ at time t. Allowing for discounting,
   $$\mathbb{E}(Y) = \int_0^n v^t\, {}_tq_x dt = \int_0^n v^t(1 - {}_tp_x)dt = \int_0^n v^t dt - \int_0^n v^t\, {}_tp_x dt = \bar{a}_{\overline{n}|} - \bar{a}_{x:\overline{n}|}.$$

   (c) The term $\bar{a}_{\overline{n}|}$ values a benefit payable continuously for n years, and the term $\bar{a}_{x:\overline{n}|}$ values a benefit payable continuously for n years provided that (x) is alive. The difference between these two terms must therefore value a benefit payable continuously for n years provided that (x) is not alive.

5. First, as
   $$\ddot{a}_{40:\overline{15}|} = 1 + v\, p_{40} + v^2\, {}_2p_{40} + \cdots + v^{14}\, {}_{14}p_{40}$$
   and
   $$a_{40:\overline{15}|} = v\, p_{40} + v^2\, {}_2p_{40} + \cdots + v^{14}\, {}_{14}p_{40} + v^{15}\, {}_{15}p_{40},$$
   subtracting gives
   $$\ddot{a}_{40:\overline{15}|} - a_{40:\overline{15}|} = 1 - v^{15}\, {}_{15}p_{40}.$$
   Rearranging and substituting gives
   $$v^{15} = \frac{1 - \ddot{a}_{40:\overline{15}|} + a_{40:\overline{15}|}}{{}_{15}p_{40}} = \frac{1 - 9.615 + 9.000}{0.9227} = 0.417254.$$
   Hence $i = (0.417254)^{-1/15} - 1 = 0.06000$ per annum.

6. Rearranging the recursive relationship, $a_x = v\, p_x(1 + a_{x+1})$,
   $$p_x = \frac{(1+i)a_x}{1 + a_{x+1}}.$$
   Substituting for x, the interest rate $i = 0.04$ and the annuity values,
   $$p_{50} = \frac{(1+i)a_{50}}{1 + a_{51}} = \frac{1.04(12.545)}{13.520} = 0.96500 \quad \text{and} \quad p_{51} = \frac{(1+i)a_{51}}{1 + a_{52}} = \frac{1.04(12.520)}{13.620} = 0.95601.$$
   Hence ${}_2p_{50} = p_{50}\, p_{51} = (0.96500)(0.95601) = 0.92255$.

7. As $A^1_{x:\overline{10|}} = A_{x:\overline{10|}} - v^{10}\,{}_{10}p_x = A_{x:\overline{10|}} - 0.375$,

$$A_{x:\overline{10|}} = 1 - d\,\ddot{a}_{x:\overline{10|}} = 1 - d(\ddot{a}_x - {}_{10|}\ddot{a}_x) = 1 - (1 - 0.94)(10.000 - 4.000) = 0.64,$$

$A^1_{x:\overline{10|}} = 0.64 - 0.375 = 0.2650.$

8. The select life table is

x	$\ell_{[x]}$	$\ell_{[x]+1}$	ℓ_{x+2}	$x+2$
50	100 557	100 455	100 320	52
51	100 401	100 284	100 134	53
52	100 221	100 095	99 927	54
53	100 020	99 882	99 693	55
54	99 795	99 639	99 429	56

The interest rate is $i = 3\%$ per annum effective.

(a)

$$\begin{aligned}\ddot{a}_{[50]:\overline{4|}} =& \frac{\ell_{[50]} + \ell_{[50]+1}v + \ell_{52}v^2 + \ell_{53}v^3}{\ell_{[50]}} \\ =& \frac{100\,557 + 100\,455(1.03)^{-1} + 100\,320(1.03)^{-2} + 100\,134(1.03)^{-3}}{100\,557} \\ =& 3.82156.\end{aligned}$$

(b)

$$\begin{aligned}a_{[50]+1:\overline{4|}} =& \frac{\ell_{52}v + \ell_{53}v^2 + \ell_{54}v^3 + \ell_{55}v^4}{\ell_{[50]+1}} \\ =& \frac{100\,320v + 100\,134(1.03)^{-2} + 99\,927(1.03)^{-3} + 99\,693(1.03)^{-4}}{100\,455} \\ =& 3.70123.\end{aligned}$$

(c)

$$\begin{aligned}(Ia)_{[50]:\overline{4|}} =& \frac{\ell_{[50]+1}v + 2\ell_{52}v^2 + 3\ell_{53}v^3 + 4\ell_{54}v^4}{\ell_{[50]}} \\ =& \frac{100\,455(1.03)^{-1} + 2(100\,320)(1.03)^{-2} + 3(100\,134)(1.03)^{-3} + 4(99\,927)(1.03)^{-4}}{100\,557} \\ =& 9.11620.\end{aligned}$$

(d)

$$\begin{aligned}(IA)_{[50]:\overline{4|}} =& \frac{d_{[50]}v + 2d_{[50]+1}v^2 + 3d_{52}v^3 + 4d_{53}v^4 + 4\ell_{54}v^4}{\ell_{[50]}} \\ =& \frac{102(1.03)^{-1} + 2(135)(1.03)^{-2} + 3(186)(1.03)^{-3} + 4(100\,134)(1.03)^{-4}}{100\,557} \\ =& 3.54759.\end{aligned}$$

(e) The probability is

$$\begin{aligned}\mathbb{P}\left[\ddot{a}_{\overline{K_{[50]}+1|}} < 5.0\right] = \mathbb{P}\left[\frac{1-v^{K_{[50]}+1}}{1-v} < 5.0\right] &= \mathbb{P}\left[K_{[50]} < \frac{\ln\left(1-5.0(1-v)\right)}{\ln(v)} - 1\right] \\ &= \mathbb{P}\left[K_{[50]} < 4.324\right] \\ &= \mathbb{P}\left[K_{[50]} \leq 4\right],\end{aligned}$$

where the last line follows as $K_{[50]}$ takes only integer values. The probability $\mathbb{P}\left[K_{[50]} \leq 4\right]$ means that ([50]) lives at most 4 complete years. In other words, that ([50]) dies before living 5 complete years from age [50]. In actuarial notation,

$$\mathbb{P}\left[K_{[50]} \leq 4\right] = {}_5q_{[50]} = 1 - \frac{\ell_{55}}{\ell_{[50]}} = 1 - \frac{99\,693}{100\,557} = 0.00859.$$